
Algorithmen und Wahrscheinlichkeit

Exercise Session 5

Exercise 1 – *Balls and Bins*

Let m, n be natural numbers. We consider the following process, which can be used, for example, to analyze the utilization of n servers to which m requests are assigned. We have m balls $b_1, \dots, b_m$ and n bins $k_1, \dots, k_n$. Each ball is thrown into a (uniformly) random bin independently of all other balls.

For the exercises below, you should find expressions that are as simple as possible (which may, of course, depend on m, n). In particular, your solution should not contain any summation signs.

- (a) Describe a probability space that describes this process.
- (b) What is the probability that the first basket k_1 is empty after all balls have been thrown?
- (c) Let X be the number of baskets that are empty after the process has ended. Determine $\mathbb{E}[X]$.
- (d) Let Y be the number of balls that land in the first basket k_1 . Determine $\mathbb{E}[Y]$.
- (e) What is the probability that the first two baskets k_1, k_2 are both empty (after all balls have been thrown)?